

SMART HIVE MONITOR: BILL OF MATERIALS (BOM) & COMPONENT REPORT

Phase 2 Prototype & Dev Kit | IEEE HardwAire Challenge

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Fx Rate: 1 USD = Rs. 95.87 INR (July 2026)

1. Hardware Sensor Node & Gateway Unit Component BOM (Verified Supplier Link Costs)

Component & Spec	Qty	Use Case / Purpose in Project	Verified Supplier Link (Ships to India)	Price (\$)	Price (Rs.)
Wio-E5 (STM32WLE5JC) Module LoRa SoC (Cortex-M4, 256k Flash)	1	Executes state machine (Wake-Sample-Infer-Act-Sleep), runs on-device decision tree classifier, drives LoRa radio.	Robu.in https://robu.in/product/seedstudio-wio-e5-lorawan-module-stm32wle5jc-embedded-arm-cortex-m4/	\$11.18	Rs. 1,072.00
Wire Antenna 868MHz Quarter-wave 86mm solid wire	1	Radiates LoRa alert packets from node to gateway; validates target >=4km LoS range KPI.	Bench stock (No external link needed)	\$0.05	Rs. 4.79
DS18B20 Temp Sensor 1-Wire digital sensor, TO-92	3	2 units measure brood-nest temperature, 1 unit measures ambient. Generates delta-T primary input for classifier.	LCSC https://www.lcsc.com/product-detail/Temperature-Sensors_XBLW-DS18B20-XBLW_C18723599.html	\$2.55	Rs. 244.47
ICS-43434 MEMS Mic I2S digital audio output microphone	1	Captures hive acoustics (FFT energy 200-400Hz) as independent cross-check signal to reduce false alarms.	DigiKey India https://www.digikey.in/en/products/detail/iknowles/ICS-43434/5180424	\$3.27	Rs. 313.49
TP4056 LiPo Charge Module 1A lithium charging module with protection	1	Manages charging LiPo cell from solar panel with over-charge and over-discharge safety path.	MakerBazar India https://makerbazar.in/products/tp4056-1a-li-ion-lithium-battery-charging-module	\$0.20	Rs. 19.17
TPS7A02 LDO Regulator Ultra-low-IQ LDO regulator (3.3V, 200mA)	1	Steps battery voltage (3.7-4.2V) down to clean 3.3V for MCU/sensors. Prevents MCU overvoltage damage.	DigiKey India https://www.digikey.in/en/products/detail/texas-instruments/TPS7A0233DBVR/16181965	\$0.55	Rs. 52.73
1000mAh 3.7V LiPo Cell 503040 slim form factor	1	Primary energy store powering node baseline runtime (>=18 months runtime with solar assist).	LCSC / local (Standard bench component)	\$1.50	Rs. 143.81
1W 6V Mini Solar Panel 60x55mm polycrystalline solar cell	1	Trickle-charges LiPo via charger under outdoor sunlight, enabling 6-month unattended operation.	LCSC / local (Standard component)	\$0.80	Rs. 76.70
Nylon Sleeveing + Spray Mesh tubing + conformal coat spray	1	Protects sensor cables from hive propolis encasement and shields PCB from internal hive moisture.	Local / bench (Consumables stock)	\$0.35	Rs. 33.55
Custom PCB (2-Layer) 60x50x1.6mm FR4, 50-ohm RF trace match	1	Main printed circuit board carrying all components with matched RF traces per ST AN5407.	JLCPCB https://jlcpcb.com/	\$0.30	Rs. 28.76
Wio-E5 Mini Dev Board Gateway Node Receiver board + USB interface	1	Physical gateway receiver board to collect and decode node transmissions and send alerts to dashboard.	Robu.in https://robu.in/product/seedstudio-wio-e5-mini-dev-board-lorawan/	\$24.36	Rs. 2,335.00
Passives & Connectors Decoupling caps, JST headers, pull-ups	1	1-Wire bus pull-up, IC decoupling capacitors, and JST battery/sensor headers.	Bench stock (Resistors & capacitors kit)	\$0.25	Rs. 23.97
SUBTOTAL - HARDWARE NODE & RECEIVER	12	Complete single-node prototype and gateway receiver hardware set.	Direct delivery to Rashtriya Raksha University.	\$45.36	Rs. 4,348.65

2. IEEE HART Development, Calibration & RF Validation Toolkit

Equipment / Item	Qty	Use Case & Experimental Purpose	Official Vendor Site & Manual Links	Price (\$)	Price (Rs.)
NanoVNA H4 Vector Network Analyzer	1	Antenna tuning, impedance matching, RF filter characterization, and S11 return loss validation to back up the Coverage Range KPI with physical measurement data.	Official site & manual: https://nanovna.com/?page_id=21	\$67.79	Rs. 6,499.00
tinySA Basic Spectrum Analyzer	1	Enables real-time LoRa transmission signal visualization at 868MHz to confirm TX frequency/power and check for local channel noise.	Wiki documentation: https://www.tinysa.org/wiki/	\$50.00	Rs. 4,793.00
ADALM2000 Module Oscilloscope + logic analyzer lab	1	Mixed-signal debugging: validates I2S audio interface timing, 1-Wire sensor bus timing, and power rail transient performance.	Analog Devices Wiki: https://wiki.analog.com/university/tools/m2k	\$99.00	Rs. 9,491.13
SUBTOTAL - IEEE HART TOOLKIT & DEV TOOLS	3	Complete RF test kit & bring-up development tools allowance.	Official Vendor Links.	\$216.79	Rs. 20,783.13

3. Requested Project Budget Breakdown (\$1,000.00 USD Funding Target / Cap)

Budget Category / Financial Pillar	Amount (USD)	Amount (INR)	Itemized Justification & Grant Purpose
Hardware Sensor Node & Gateway Unit BOM	\$45.36	Rs. 4,348.65	Core single-node sensor prototype (\$21.00) + gateway receiver board (\$24.36) to assemble the physical hardware nodes.
IEEE HART RF Test Toolkit (3 Units)	\$216.79	Rs. 20,783.13	NanoVNA H4 (\$67.79), tinySA Basic (\$50.00), ADALM2000 (\$99.00) - RF bring-up tools required for validation.
Field Deployment Rigs, Mounts & Weatherproofing	\$200.00	Rs. 19,174.00	Outdoor hive mounting rigs, IP67 enclosure seals, thermal insulation & field assembly tools.
Travel Expenses, DDP Shipping & Field Logistics	\$200.00	Rs. 19,174.00	Field site travel, express courier shipping to RRU, GST taxes, customs clearance & transit logistics.
TOTAL REQUESTED GRANT BUDGET	\$662.15	Rs. 63,479.78	Official total grant funding requested from IEEE HART Challenge Platform.
Maximum Platform Budget Ceiling	\$1,000.00	Rs. 95,870.00	Maximum grant funding cap allowed under IEEE HART Challenge rules.

NOTE:- Shipping costs are excluded from this Bill of Materials (BOM).

This BOM is subject to change based on component availability.

Parts will be procured from the most readily available sources, prioritizing availability over cost. Spare components will be purchased in advance. If any individual part or the entire device is damaged during assembly, replacements will be procured.

All leftover unspent funds will be returned to RRU. Any other expenses incurred by team members will be returned/settled upon grant reimbursement.

ACKNOWLEDGEMENT:- All financial decisions will be made collaboratively by the entire team.